CS4116 Design Document

Fields & Futures

Group 2 - Members

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Introduction:

Our dating platform, Fields & Futures, is designed to support meaningful, long-term relationships within rural communities. In a world where many dating apps prioritise fast swiping and casual interactions, Fields & Futures focuses on intentional dating rooted in shared values and lifestyle compatibility. The platform allows users to create detailed profiles that expertly reflect their personal interests. With a focus on “intentional dating”, Fields & Futures is one of the first dating websites to encourage users to give each potential match through and genuine consideration. Users can search for others based on profile information such as age, gender, location, and selected interests, with a particular emphasis on rural living and compatibility of long-term intentions. Registered users can express interest in one another. When two users mutually indicate interest, a match is formed, and they are able to communicate through our platform’s internal messaging system. Fields & Futures aims to provide a slower, more thoughtful alternative to mainstream dating platforms by prioritising authenticity and genuine future planning.

High-Level Functionality

On Fields & Futures, there are two types of users, admin users and basic users. The basic users can create their account by inputting an email, username, and a password. After creating an account, the user is brought to the user profile page to add their profile pic, bio, personal details, interests, and search settings to their user account. After filling in these fields, the user is directed to the main page where they can start matching with other users based on their interests. On this main page, the user has the option to view the website guidelines, chat with users they have matched with, and to edit their profile further.

Users are matched with other users based off of a "Compatibility Score" determined by the matching interests between users, preferences, and other criteria provided by the users. The functionality of the admin users is to view all existing users, edit user profiles, ban or suspend users and activate or deactivate accounts. To achieve this, the admins can view and act on reports submitted by the users.

Required Functionality

User Registration & Profile

- Create an account and log in securely
- Create and edit a personal profile
- Provide details such as description, interests, preferences, and basic information

Passwords **will not be stored in plain text**

Browsing & Searching

- Browse other users
- Search using multiple criteria
- Apply filters and sorting

Search functionality will be implemented using database queries as opposed to manually filtering in PHP

Matching & Connecting

- Express interest in another user
- Form a connection only if both users indicate interest
- View existing matches
- Complain about harassment from users
- Block other users

Messaging

Matched users must be able to communicate through an **internal messaging system**. The system should identify and prevent users from sending phone numbers through the messaging system - we want to keep users on our site as much as possible!

Administrative Controls

- Ban or suspend users
- Remove inappropriate content
- Edit user profile

Security

- Password hashing
- Secure session handling
- Input validation and sanitisation
- Protection against basic SQL injection
- No fatal runtime errors
- No hard-coded credentials
- Clean and consistent navigation
- Logical file structure

Additional Functionality

Intentional Dating

- Including guidelines on how to do Intentional dating
- Setting a limit for number of matches at a time
- Setting a limit for the number of messages between two accounts

Compatibility score

- Based on interests, location & sexuality show relevant profiles

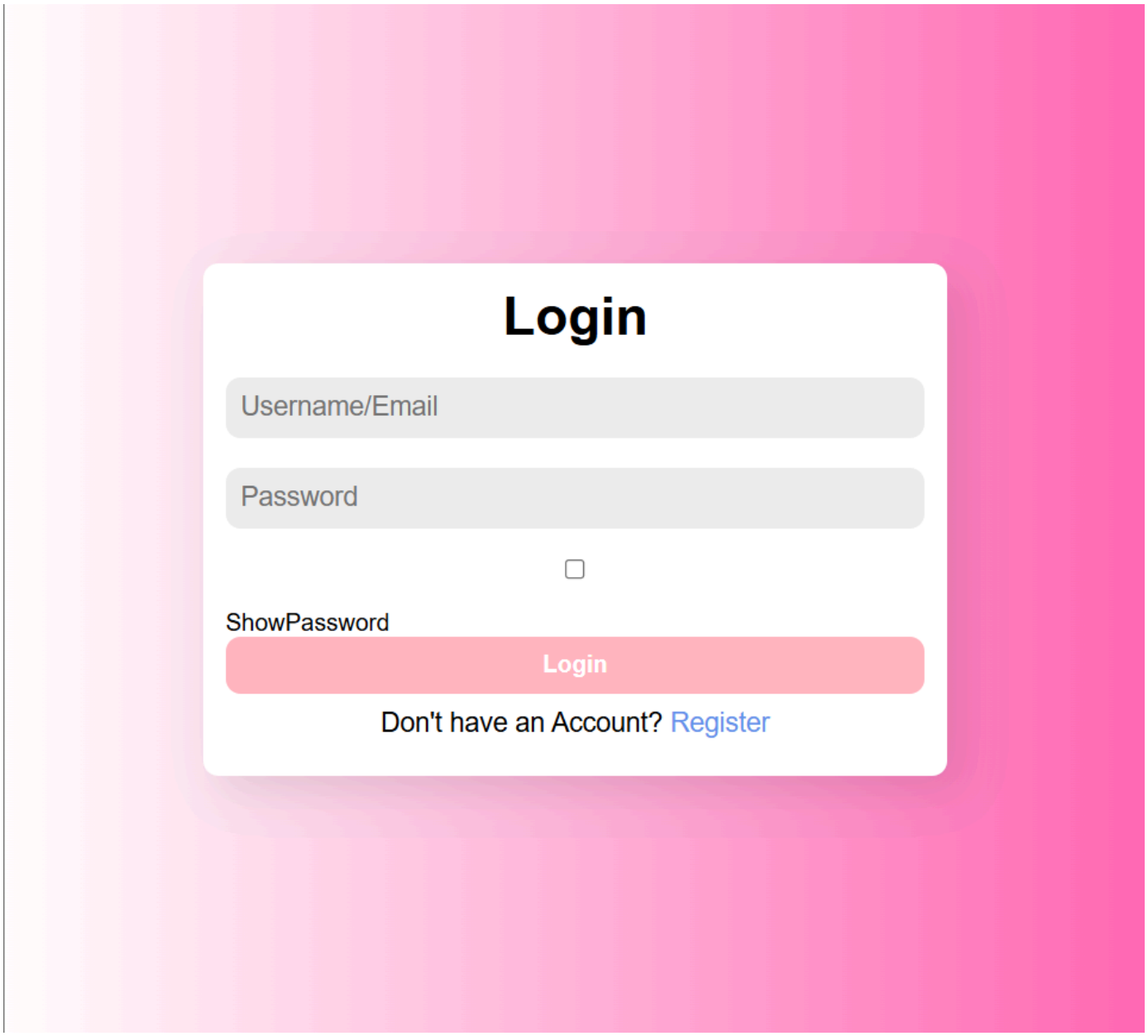
Ice-Breakers

- Prompt users with Ice-Breakers
- When reaching message limit prompt users with date ideas

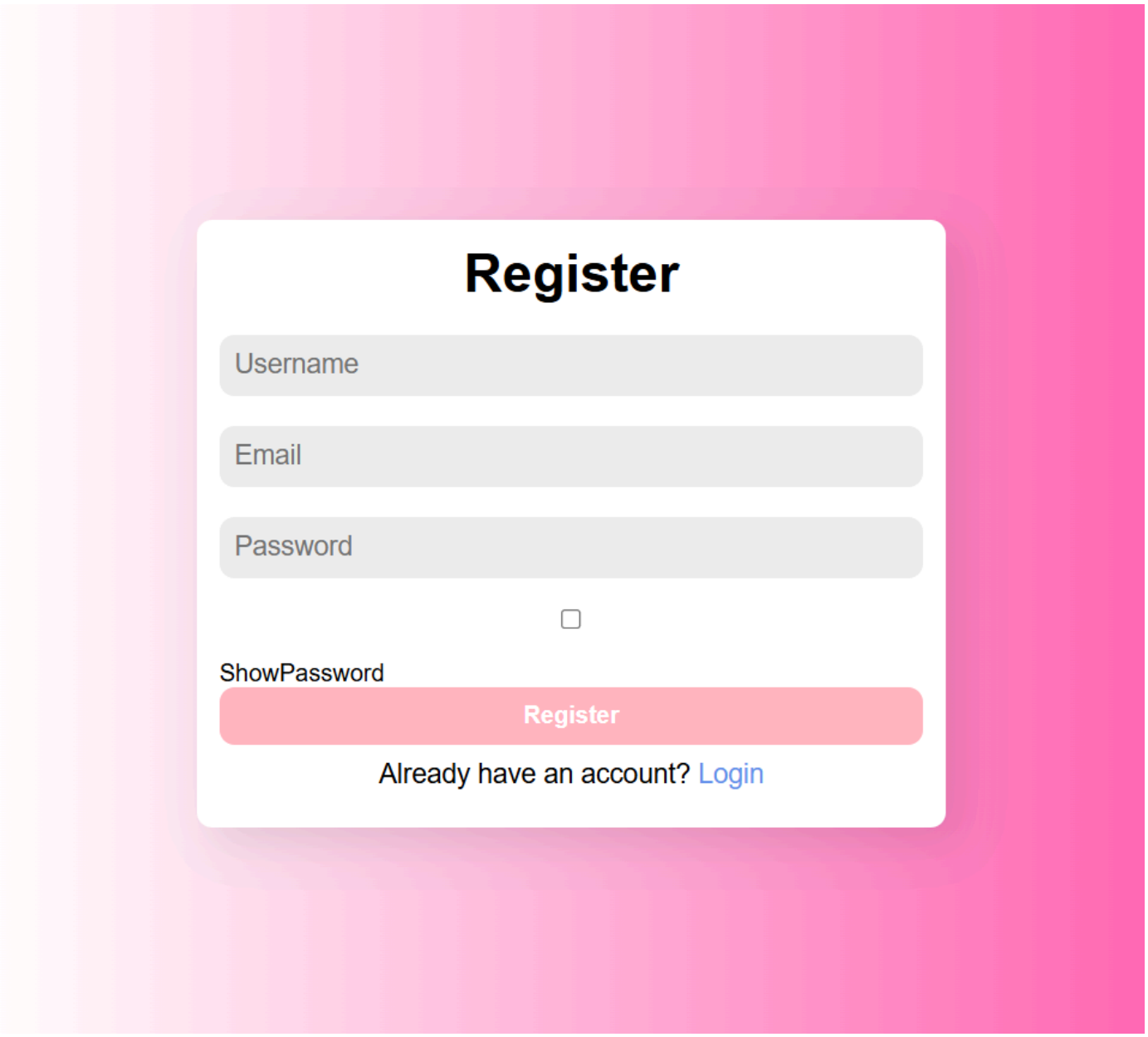
WebPage Mockups

Google-based UI mockups

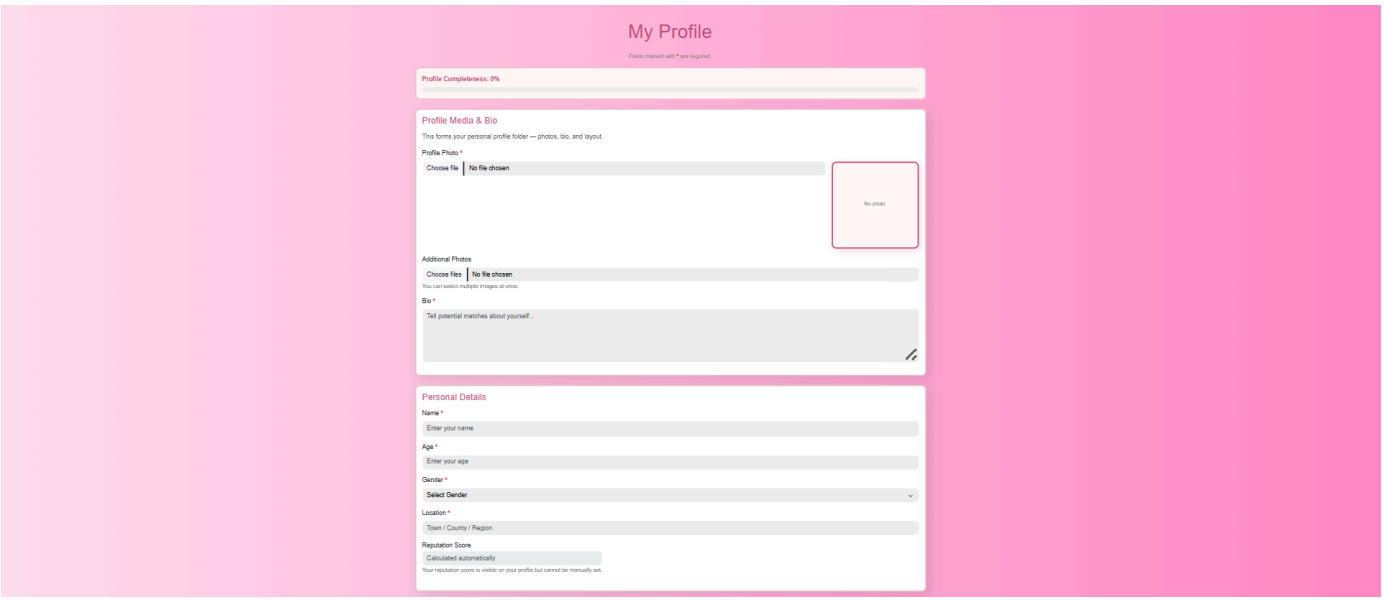
Login page

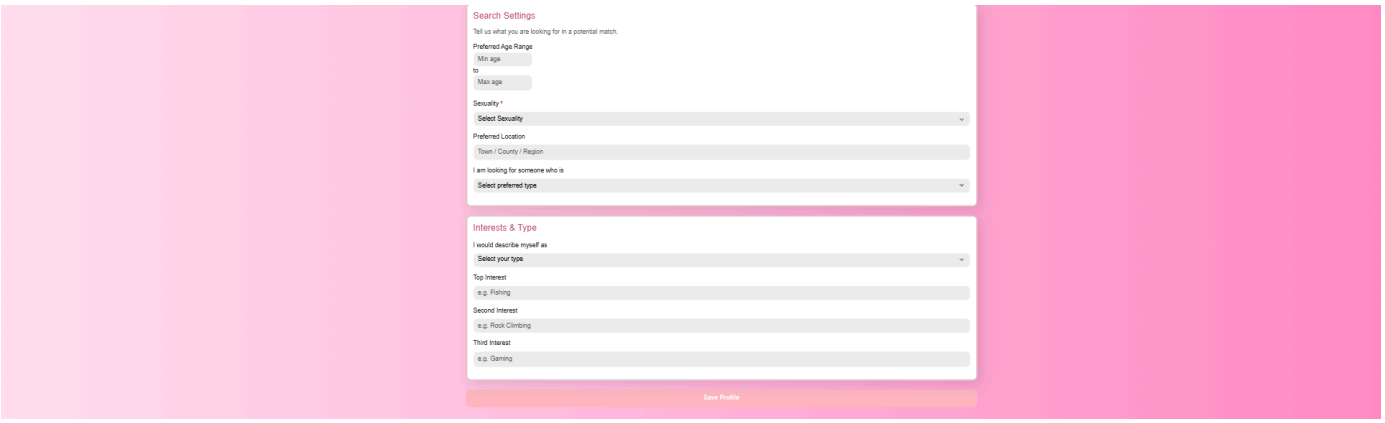


Register page

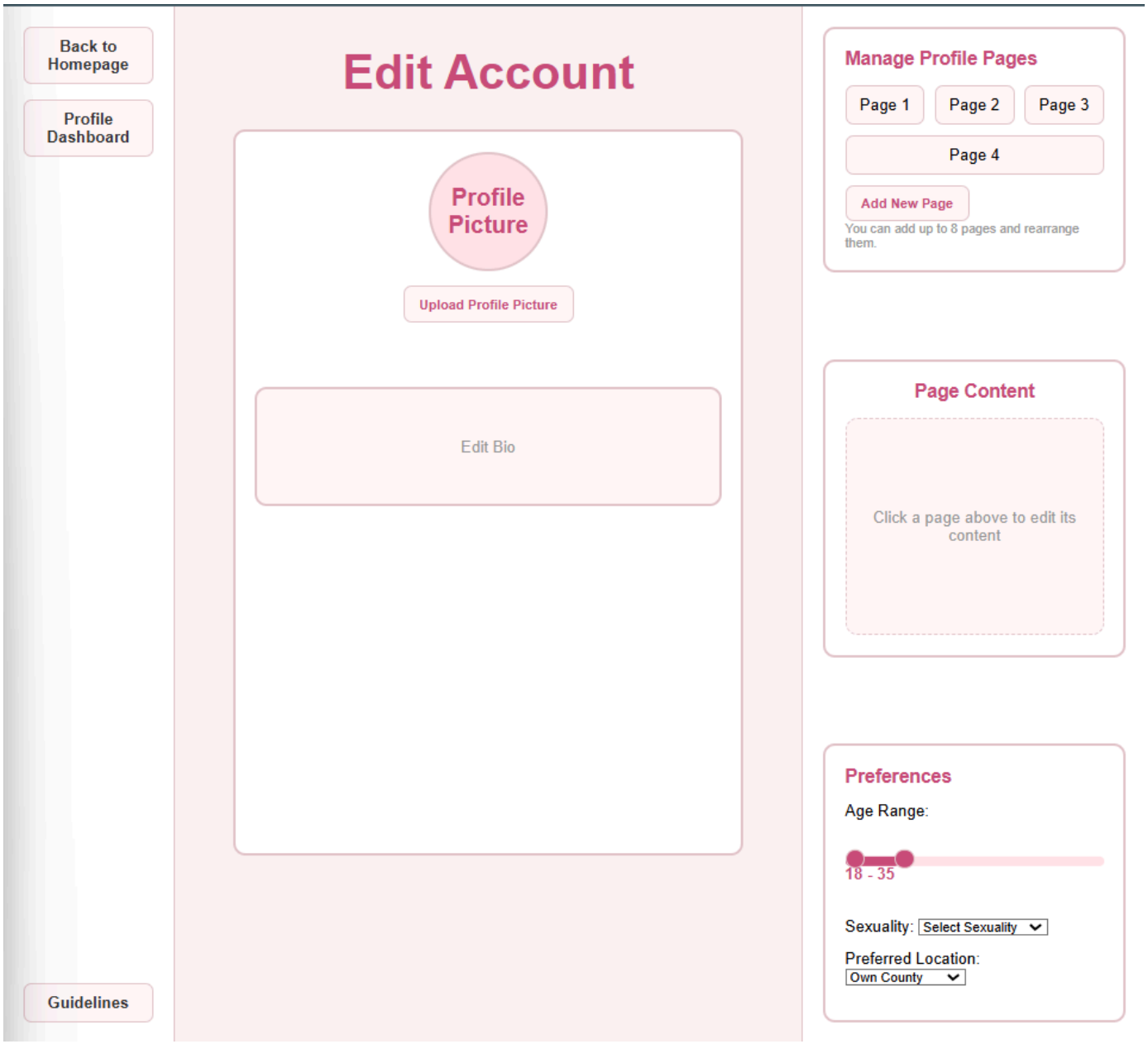


My Profile page section





Edit Account page



Guidelines page

Edit Account

Profile Dashboard

Guidelines

Recommended User Guidelines

Welcome (Name of User) to our website. We know you are eager to meet your match and to help you on this journey, we have made some guidelines on how we intend for you to use our website.

- Guideline 1
- Guideline 2
- Guideline 3

By following the guidelines above, it will enhance your chances of finding the love of your life. Our goal is to improve yours, and your matches user experience. We don't tolerate any abuse and an account ban will take place if any of the below are violated!

- Guideline 1
- Guideline 2
- Guideline 3

5

Messages

Chat with Emma

Chat with Liam

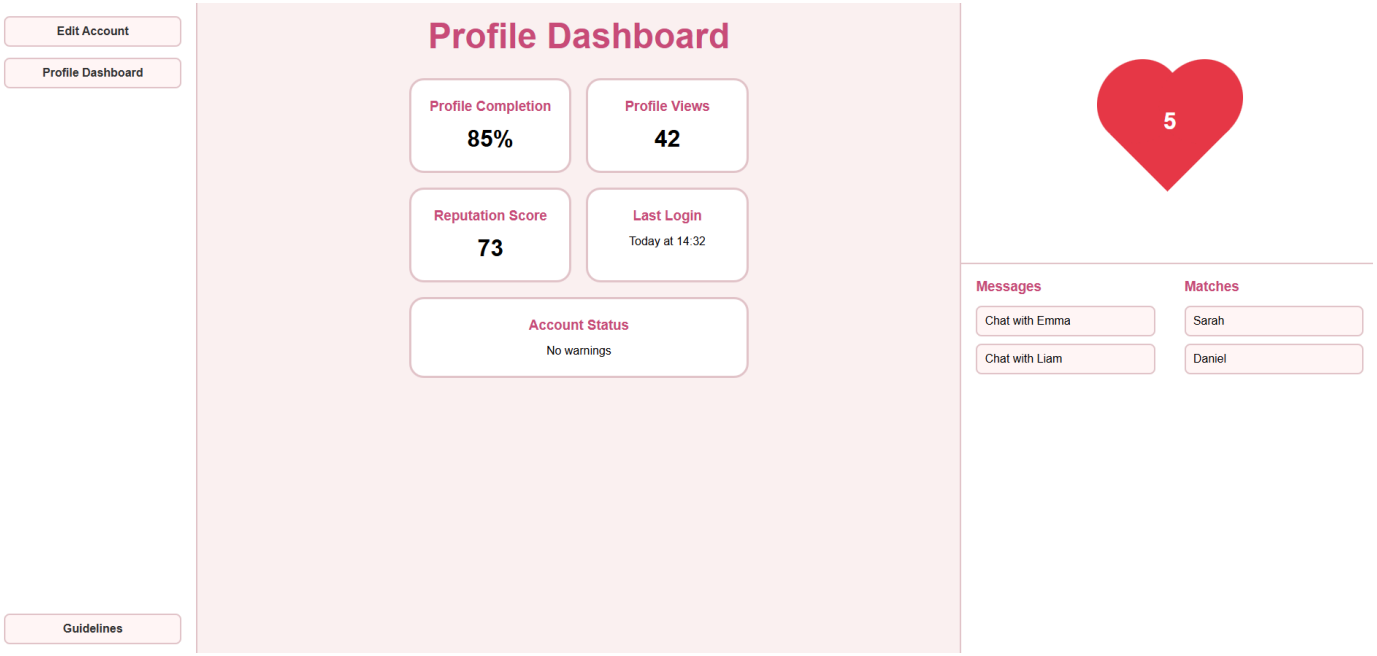
Matches

Sarah

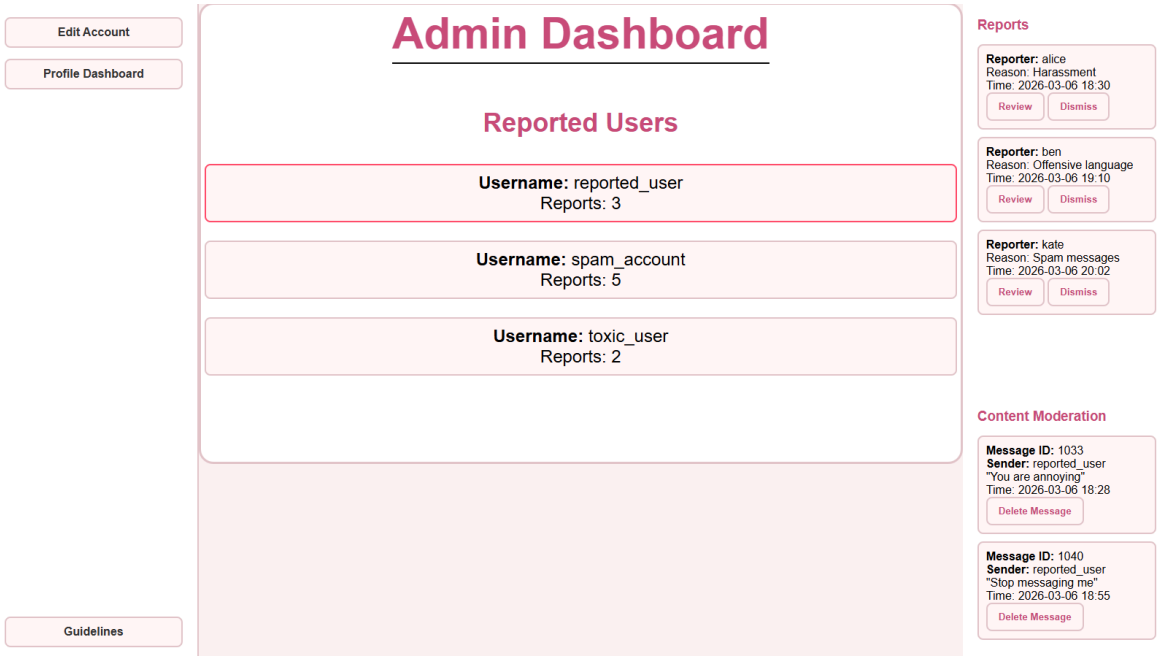
Daniel

Guidelines

Profile Overview page



Admin Dashboard page



Database Tables

Dictionary:

PK - Primary Key

FK - Foreign Key

User set-up Table

This table serves as the foundational database structure for the dating platform, handling account registration, personal profile details,and secure login credentials. Additionally, it includes crucial fields like role and ban status to support the required administrative controls for managing and moderating the user base.

Field	Type	KeyType
user_id	INT	PK
username	VARCHAR	UNIQUE
email	VARCHAR	UNIQUE
password_hash	VARCHAR	
role	ENUM('user','admin')	
gender	ENUM('male','female','non_binary','other')	
date_of_birth	DATE	
country	VARCHAR	
bio	TEXT	
reputation_score	INT	
is_active	BOOLEAN	
is_banned	BOOLEAN	
created_at	TIMESTAMP	
last_login_at	TIMESTAMP	
deleted_at	TIMESTAMP (nullable)	
profile_completed	BOOLEAN	

Permissible Interests Table

This table acts as a standardised reference dictionary for all the predefined hobbies and passions users can add to their personal profiles. By restricting selections to a consistent set of unique tags rather than free-text entry, it keeps our database clean and makes the required feature of searching and filtering by interests much more accurate and reliable.

Field	Type	KeyType
interest_id	INT	PK
interest_name	VARCHAR	UNIQUE

User interest Table

This table links individual members directly to their chosen hobbies from the permissible interests list. By matching a user_id with an interest_id, this junction table efficiently tracks the specific interests of our users, which directly powers our platform's ability to search and filter profiles based on shared passions.

Field	Type	KeyType
user_id	INT	PK, FK → users.user_id
interest_id	INT	PK, FK → interests.interest_id

Preferences Table

This table stores the specific dating criteria for each individual in our database, fulfilling the requirement to capture user preferences. By keeping track of requirements like preferred age ranges and county, it allows our system to proactively suggest compatible matches. Furthermore, it provides the exact data structure we need to power the required search and filtering functionality.

Field	Type	KeyType
user_id	INT	PK, FK → users.user_id
min_age	INT	
max_age	INT	
prefers_male	BOOLEAN	
prefers_female	BOOLEAN	
prefers_non_binary	BOOLEAN	
prefers_other	BOOLEAN	
preferred_county	VARCHAR (Nullable)	

Likes Table

This table records every instance where a member of our database expresses interest in another user. By tracking exactly who likes whom, it allows our platform to manage user connection to be used in the matching process.

Field	Type	KeyType
liker_id	INT	PK, FK → users.user_id
liked_id	INT	PK, FK → users.user_id
created_at	TIMESTAMP	

Matches Table

This table records the established connections within our database when two users successfully express mutual interest. By tracking the exact status of a relationship (such as active, unmatched, or blocked), it enables our platform to manage clear state transitions and allows users to securely view their existing matches. This table also includes a compatibility score associated with each match to help us proactively evaluate and suggest high-quality pairings based on user preferences.

Field	Type	KeyType
match_id	INT	PK
user1_id	INT	FK → users.user_id
user2_id	INT	FK → users.user_id
status	ENUM('active','unmatched','blocked')	
created_at	TIMESTAMP	
match_compatability_score	INT	
updated_at	TIMESTAMP	
deleted_at	TIMESTAMP (nullable)	

Messages Table

This table powers the internal messaging system in our database, securely facilitating communication between users who have successfully matched. By linking the message text to a specific match id, it ensures conversations only happen between mutual connections and that the timestamps accurately reflect the time of the message.

Field	Type	KeyType
message_id	INT	PK
match_id	INT	FK → matches.match_id
sender_id	INT	FK → users.user_id
message_text	TEXT	
is_read	BOOLEAN	
created_at	TIMESTAMP	
deleted_at	TIMESTAMP (nullable)	

Blocks Table

This table is for keeping track of exactly who has blocked who in our database. It's necessary so that we can easily stop unwanted interactions and make sure a blocked person can't see or message the user who blocked them. Basically, it handles the privacy side of things to keep the platform safe and comfortable for everyone.

Field	Type	KeyType
block_id	INT	PK
blocker_id	INT	FK → users.user_id
blocked_id	INT	FK → users.user_id
created_at	TIMESTAMP	

Reports Table

This table acts as the moderation hub in our database, letting users officially complain about harassment or inappropriate behaviour from others. It's necessary so that our admin team can track the status of each issue, review specific messages, and decide if someone needs to be banned or suspended. Ultimately, it organises all these reports so we can effectively moderate the platform.

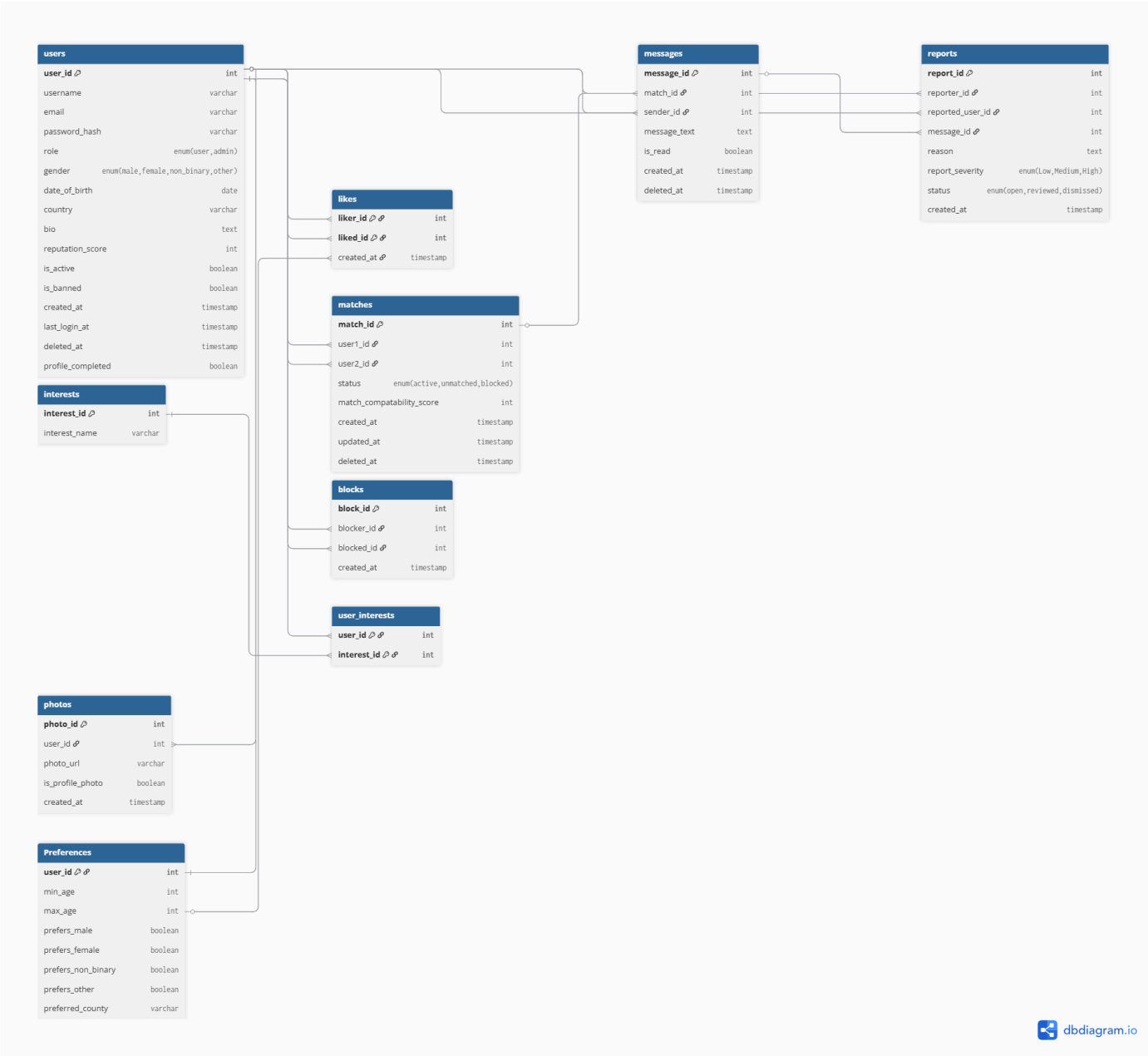
Field	Type	KeyType
report_id	INT	PK
reporter_id	INT	FK → users.user_id
reported_user_id	INT	FK → users.user_id
message_id	INT (nullable FK → messages.message_id)	
reason	TEXT	
report_severity	ENUM('Low','Medium','High')	
status	ENUM('open','reviewed','dismissed')	
created_at	TIMESTAMP	

Photos Table

This table stores the links for all the pictures our users upload to create and edit a personal profile. It's necessary in our database to link each image to the right person and explicitly flag their main display picture so people can actually see their potential matches when they browse other users.

Field	Type	KeyType
photo_id	INT	PK
user_id	INT	FK → users.user_id
photo_url	VARCHAR	
is_profile_photo	BOOLEAN	
created_at	TIMESTAMP	

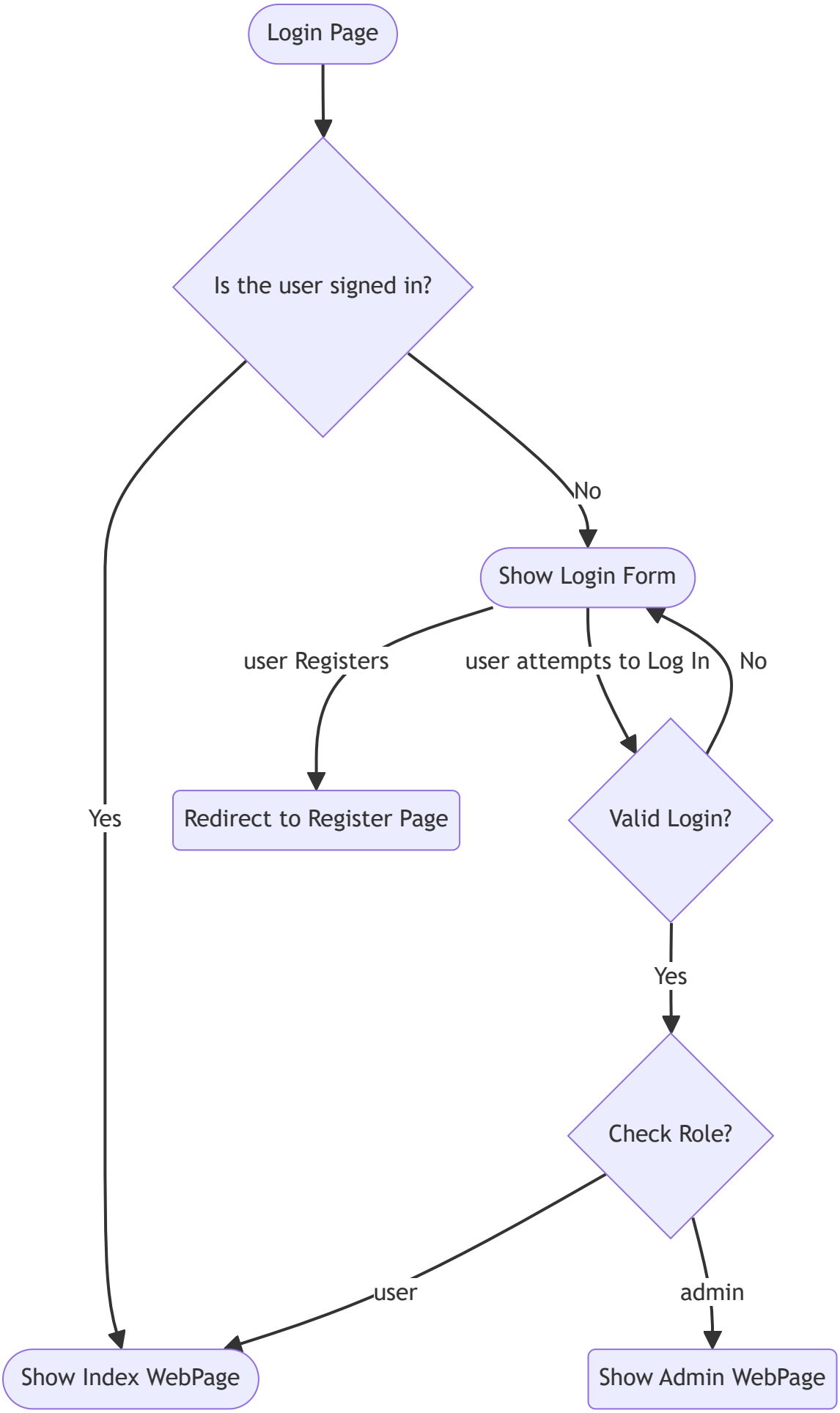
Entity Relationship Diagram



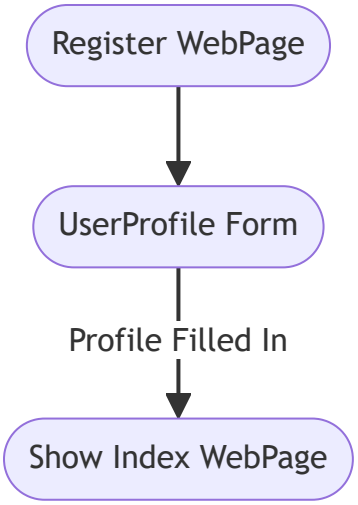
Process Chart List

User

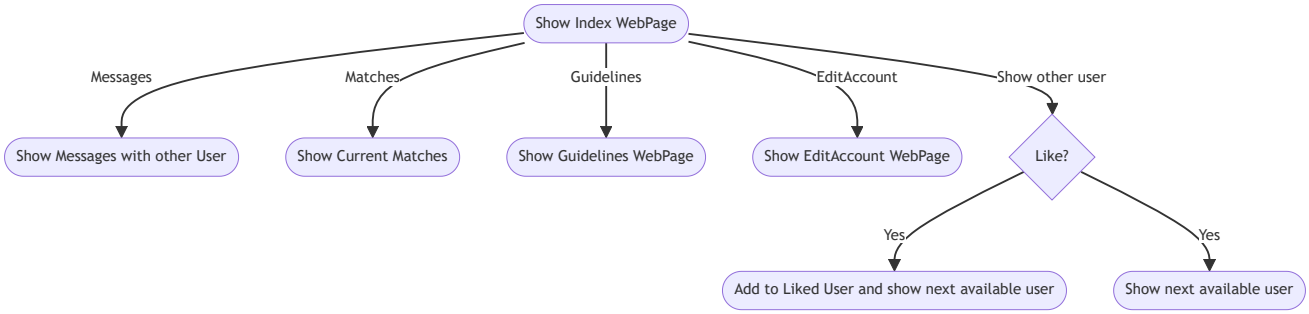
Login


Login Page
Is the user signed in?
Yes
No
Show Login Form
user Registers
user attempts to Log In
No
Redirect to Register Page
Valid Login?
Yes
Check Role?
user
admin
Show Index WebPage
Show Admin WebPage

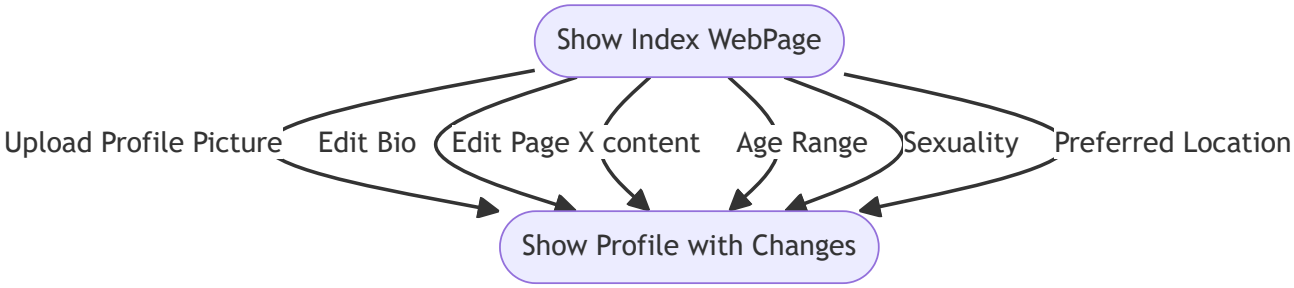
Registration



Index page

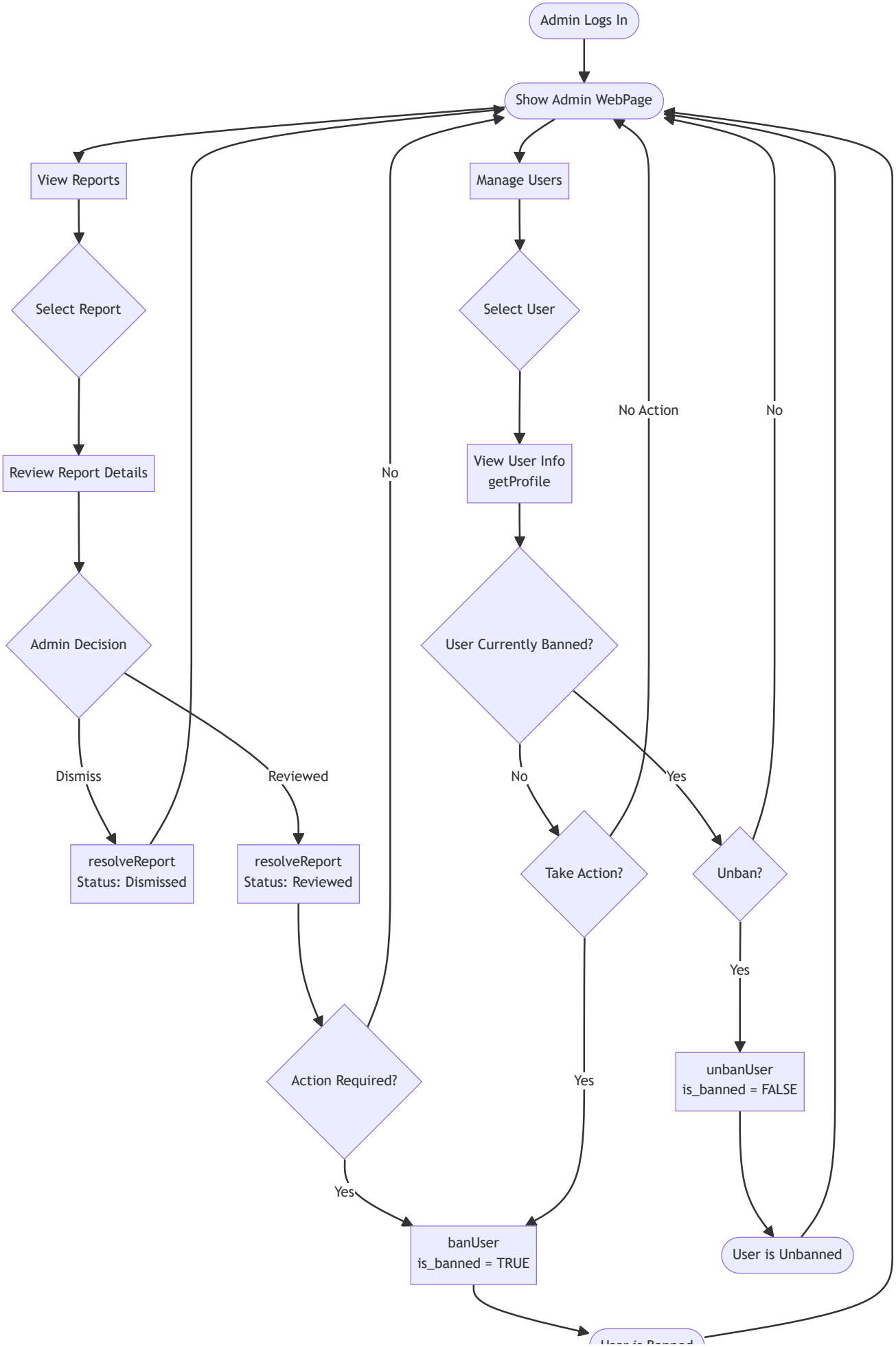


Edit Account



Admin

Admin Actions


Admin Logs In
Show Admin WebPage
View Reports
Manage Users
Select Report
Select User
Review Report Details
View User Info
getProfile
Admin Decision
User Currently Banned?
Dismiss
Reviewed
No
Yes
resolveReport
Status: Dismissed
resolveReport
Status: Reviewed
Take Action?
Unban?
No Action
No
No
Yes
Action Required?
Yes
unbanUser
is_banned = FALSE
Yes
banUser
is_banned = TRUE
User is Unbanned

User is Banned

Process Tables

Process No.	1
Title	registrationValidation
Brief Description	Validates a users credentials when registering
Inputs	Username, Email, Password
Detailed Description	Validates that an account does not already exist with the username or email a user is attempting to register with. If an account exists in the database with either of these an error pops up and the user is prompted to enter new credentials. If these credentials are not found, the createAccount process is allowed to proceed.
Outputs	userID

Process No.	2
Title	createAccount
Brief Description	Creates a new user account after successful registration
Inputs	Username, Email, Password
Detailed Description	Adds a new user to the database, assigning them a unique user_id and hashing their password.
Output	User gets prompted to login using their credentials

Process No.	3
Title	loginValidation
Brief Description	Validates a users credentials when attempting to log in
Inputs	Username/Email, Password
Detailed Description	Validates that the username/email and password entered match an existing account. Returns a boolean depending on the success of this check.
Output	Boolean True/False to indicate if login should be allowed.

Process No.	4
Title	profileSetup

Process No.	4
Brief Description	Checks if a users profile has been setup.
Inputs	Email/Username
Detailed Description	When a user is logging in, their credentials are checked to see if their account has had its profile setup completed.
Output	Boolean True/False to indicate if the user should be sent to the inital profile setup page or the websites home page.

Process No.	5
Title	loginUser
Brief Description	Logs a user into the website
Inputs	loginValidation Boolean, profileSetup Boolean
Detailed Description	Based on the values returned from processes 3 and 4, this process will either deny or allow a user to be logged in, and send them to the page relevant to them.
Output	User is either logged in or denied. If logged in they are sent to the initial profile setup if it has not yet been completed or the website homepage if it has.

Process No.	6
Title	setCompleteness
Brief Description	Sets the completeness percentage of a users profile.
Inputs	Profile Photo, Bio, Name, Age, Gender, Location, Type, Interests
Detailed Description	When filling out a profile initially, a completion percentage is present on the page which dynamically updates as the profile is filled out. Only certain fields count towards this value.
Output	Completion Percentage

Process No.	7
Title	requiredCheck
Brief Description	Checks if the required values are filled out during profile setup.
Inputs	Name, Age, Location, Bio, Profile Photo, Gender

Process No.	7
Detailed Description	If these required values are not filled out, the user is not allowed to save their profile. If they are, then the information they provided will be saved to their profile using process 8.
Output	Boolean True or False, to allow the user to save the profile onto the database

Process No.	8
Title	saveProfile
Brief Description	Saves a users profile information
Inputs	Profile Photo, Bio, Name, Age, Gender, Location, Type, Interests
Detailed Description	This takes all the inputs and saves them to the database
Output	Sends the user to their profile webpage and saves their values to the database

Process No.	9
Title	getProfile
Brief Description	Return a representation of the selected users profile
Inputs	userID
Detailed Description	This function will return a text version of the selected users profile
Output	A String with all the information about the selected users profile

Process No.	10
Title	changeEmail
Brief Description	Allows a user to change email
Inputs	Oldemail, Newemail & password
Detailed Description	This will replace the saved email for the current user
Output	Boolean True or False, To tell the user the action has been completed

Process No.	11
Title	changePassword
Brief Description	Allows a user to change password
Inputs	Email, Oldpassword & Newpassword
Detailed Description	This will replace the saved password for the current user

Process No.	11
Output	Boolean, To tell the user the action has been completed

Process No.	12
Title	setProfilePhoto
Brief Description	Allows a user to set their profile photo
Inputs	Any image format HTML accepts
Detailed Description	Let's a user provide images which will be used with their profile
Output	Boolean, To tell the user the action has been completed

Process No.	13
Title	getProfilePhoto
Brief Description	Allows a user to check the set profile photo
Inputs	userID
Detailed Description	Let's a user get the photos set to their profile photo
Output	Zip file with the image

Process No.	14
Title	setAdditionalPhotos
Brief Description	Allows a user to set their additional photos
Inputs	Any image format HTML accepts
Detailed Description	Let's a user provide images which will be used with their additonal photos section
Output	Boolean, To tell the user the action has been completed

Process No.	15
Title	getAdditionalPhotos
Brief Description	Allows a user to check the set additional photos
Inputs	userID
Detailed Description	Let's a user get the photos set to their additional photos
Output	Zip file with the images

Process No.	16
Title	setBio
Brief Description	Let's a user set their profile bio
Inputs	String
Detailed Description	A user will fill in a text field which will be saved to the database
Output	Boolean, To tell the user the action has been completed

Process No.	17
Title	getBio
Brief Description	Returns the Bio for the selected users profile
Inputs	userID
Detailed Description	Returns the String that has been saved into the database for the users' Bio
Output	String, of the attached Bio

Process No.	18
Title	setName
Brief Description	Let's a user set their name
Inputs	String
Detailed Description	Saves the users chosen name to the database
Output	Boolean, To tell the user the action has been completed

Process No.	19
Title	getName
Brief Description	Returns the Name for the selected user
Inputs	userID
Detailed Description	Returns the String that has been saved into the database for the users' Name
Output	String

Process No.	20
Title	setAge
Brief Description	Let's a user set their Age

Process No.	20
Inputs	Int
Detailed Description	Saves a users' chosen Age to the database
Output	Boolean, To tell the user the action has been completed

Process No.	21
Title	getAge
Brief Description	Returns a users' Age
Inputs	userID
Detailed Description	Returns the Int saved into the database for the selected users' age
Output	Int, Of the users' Age

Process No.	22
Title	setLocation
Brief Description	Let's a user set their Location
Inputs	String
Detailed Description	Saves a users' chosen Location to the database
Output	Boolean, To tell the user the action has been completed

Process No.	23
Title	getLocation
Brief Description	Returns a users' Location
Inputs	userID
Detailed Description	Returns the String saved into the database for the selected users' Location
Output	String, Of the users' Location

Process No.	24
Title	calculateReputation
Brief Description	Calculates and updates a users reputation
Inputs	UserID

Process No.	24
Detailed Description	Calculates the users reputation score by identifying reports against the user, and updates the recorded reputation
Output	Returns updated reputation score. Internal method never seen by the user so does not need to display anything on-screen

Process No.	25
Title	getReputation
Brief Description	Returns the users current reputation score
Inputs	Necessary for process #24. Gets the users current reputation score
Detailed Description	Retrieves the users reputation score.
Output	Returns users reputation score. Internal method never seen by the user so does not need to display anything on-screen

Process No.	26
Title	setType
Brief Description	Let's a user set their Type
Inputs	Enum
Detailed Description	Saves a users' chosen Type to the database
Output	Boolean, To tell the user the action has been completed

Process No.	27
Title	getType
Brief Description	Returns a users' Type
Inputs	userID
Detailed Description	Returns the Enum saved into the database for the selected users' Type
Output	Enum, Of the users' Type

Process No.	28
Title	setInterests
Brief Description	Let's a user set their Interests

Process No.	28
Inputs	Enum
Detailed Description	Saves a users' chosen Interests to the database
Output	Boolean, To tell the user the action has been completed

Process No.	29
Title	getInterests
Brief Description	Returns a users' Interests
Inputs	userID
Detailed Description	Returns the Enum saved into the database for the selected users' Interests
Output	Enum of the users' Type

Process No.	30
Title	newMatch
Brief Description	Creates a new match between two users
Inputs	user1_id, user2_id
Detailed Description	Creates a new entry in the matches table, assigning it a unique match_id. It is given a created_at timestamp and its status is set to active.
Output	The user is informed that a new match has been created and process 32 is called to decrement their remaining matches.

Process No.	31
Title	removeMatch
Brief Description	Changes a matches status
Inputs	user1_id, user2_id
Detailed Description	Changes a matches status from active to inactive/blocked depending on if the match was ended normally, or a user was blocked.
Output	The user is informed that the match has been removed and process 32 is called to increment their remaining matches.

Process No.	32
Title	matchesRemaining

Process No.	32
Brief Description	Sets the number of remaining matches a user has and returns the value
Inputs	user_id
Detailed Description	Checks the matches table for active matches associated with the user_id and updates the value.
Output	INT

Process No.	33
Title	sendMessage
Brief Description	Sends a message to another user
Inputs	match_id, sender_id, message_text
Detailed Description	Creates a new message in the messages table, using the senders id and the match id to identify the sender and recipient.
Output	created_at TIMESTAMP, for the message

Process No.	34
Title	getMessages
Brief Description	Gets a users messages
Inputs	match_id
Detailed Description	Returns the message history between two users by querying their match id in the messages table.
Output	message_text for each message found

Process No.	35
Title	setRead
Brief Description	Sets a messages status to read
Inputs	message_id
Detailed Description	Once a message is read by the recipient, its is_read value is updated to be TRUE and this value is returned.
Output	Boolean TRUE, to let the user know the message has been read

Process No.	36
Title	sendLike
Brief Description	Sends a like from one user to another.
Inputs	liker_id, liked_id
Detailed Description	Creates a new like in the likes table, using the liker_id and liked_id for identification. It is then given a created_at timestamp.
Output	Boolean response that the like was successful.

Process No.	37
Title	getLikes
Brief Description	Returns all the likes for a user.
Inputs	liked_id
Detailed Description	Using the liked_id, all of the likes sent to this id are counted and returned when a user wants to see how many likes they have.
Output	INT amount of likes a user has received

Process No.	38
Title	blockUser
Brief Description	Blocks another user
Inputs	blocker_id, blocked_id
Detailed Description	When a user blocks another user, a new block is created in the blocks table with a unique block_id. Both of their individual id's are stored here along with a created_at field for the block.
Output	Boolean response to inform user that they have successfully blocked the other.

Process No.	39
Title	unblockUser
Brief Description	Unblocks another user
Inputs	block_id
Detailed Description	Sets a blocks status to inactive in the blocks table using the block_id.
Output	Boolean response to inform user that they have successfully unblocked the other.

Process No.	40
Title	setActive
Brief Description	Toggles a users account status
Inputs	user_id
Detailed Description	Toggles a users account between active & deactivated by modifying the is_active field associated with their user_id.
Output	None

Process No.	41
Title	getActive
Brief Description	Gets a users account status
Inputs	user_id
Detailed Description	Returns the current status of a users account through a boolean value
Output	Boolean to represent if account is active

Process No.	42
Title	sendReport
Brief Description	Creates a report against another user
Inputs	reporter_id, reported_user_id, reason, report_severity
Detailed Description	Creates a new report in the reports table and maps it to the reporting and reported users using their user_id.
Output	Boolean to inform the user the report was successfully created.

Process No.	43
Title	getReportStatus
Brief Description	Gets a reports status
Inputs	report_id
Detailed Description	Returns the status of a report from the reports table by searching for the report_id.
Output	ENUM for status

Process No.	44
Title	resolveReport
Brief Description	Allows an admin to resolve a report.
Inputs	report_id
Detailed Description	An admin can review a report and set its status to reviewed or dismissed using buttons on the admin page.
Output	None

Process No.	45
Title	banUser
Brief Description	Allows an admin to ban a user.
Inputs	user_id
Detailed Description	An admin can enter a users id to ban them from the website, which sets their is_banned value to true.
Output	Boolean is_banned, to represent the account is now banned.

Process No.	46
Title	unbanUser
Brief Description	Allows an admin to unban a user.
Inputs	user_id
Detailed Description	An admin can enter a users id to unban them from the website, which sets their is_banned value to false.
Output	Boolean is_banned, to represent the account has been unbanned.

Process No.	47
Title	setGender
Brief Description	Sets a users gender
Inputs	user_id, gender
Detailed Description	Sets the gender for a users account in the user set-up table to be whatever gender the user chose.
Output	Boolean to inform the user the action was successful.

Process No.	48
Title	getGender
Brief Description	Returns a users gender
Inputs	user_id
Detailed Description	Returns a users gender from the user set-up table by searching for their user_id
Output	ENUM for gender